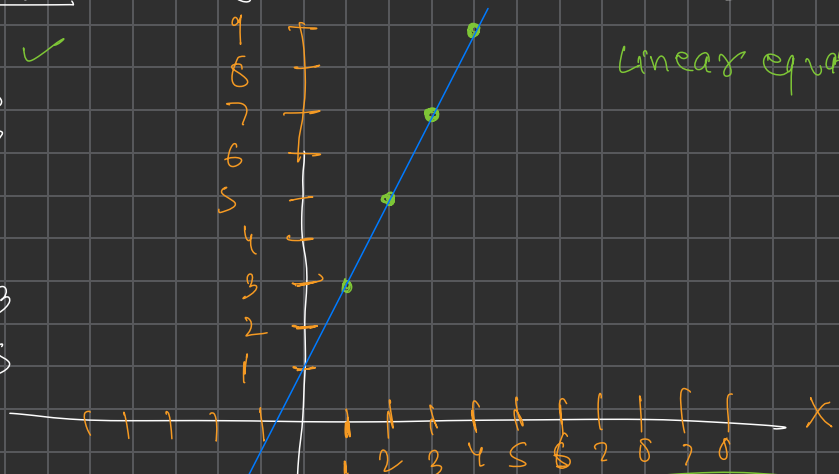
(Studhours) X	Y (Marks)
1	3 ✓
2	5
3	7
4	9
5	11
6	13
7	15

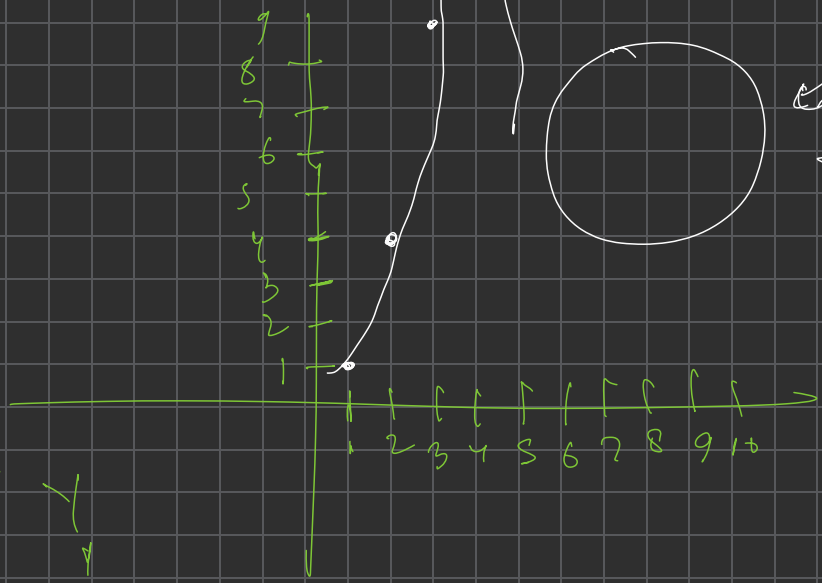
$y = 2x+1$ $[Y = w_1 x + \theta]$

Linear equation



$[5x+3y \neq Y = \text{output}]$

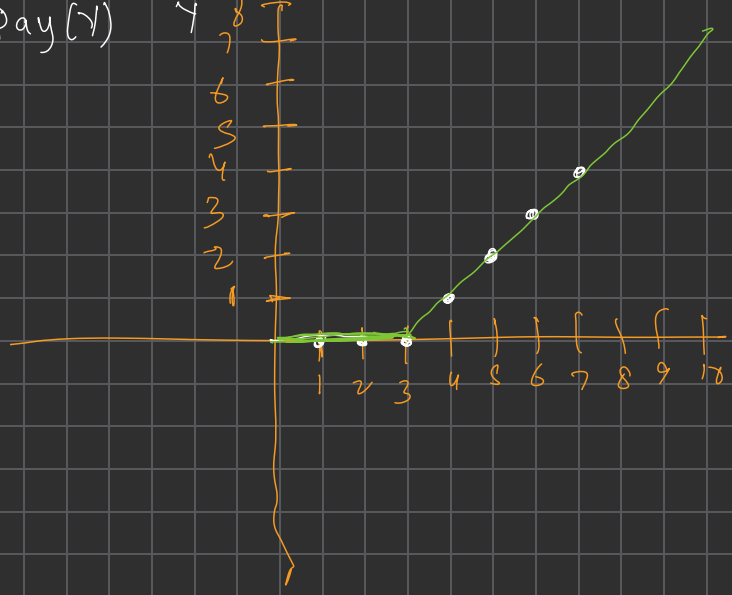
$\boxed{y = x^2}$



circle

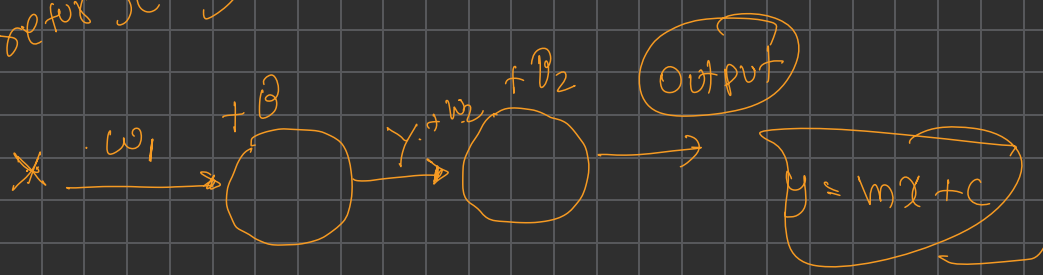
X	Y
1	1
2	4
3	9
4	16
5	25

(x) Hour	Extra Pay (Y)
1	0
2	0
3	0
4	1
5	2
6	3
7	4
8	5




```
if (x <= 3)
    return 0;
else
    return x - 3
```

$y = x^2$

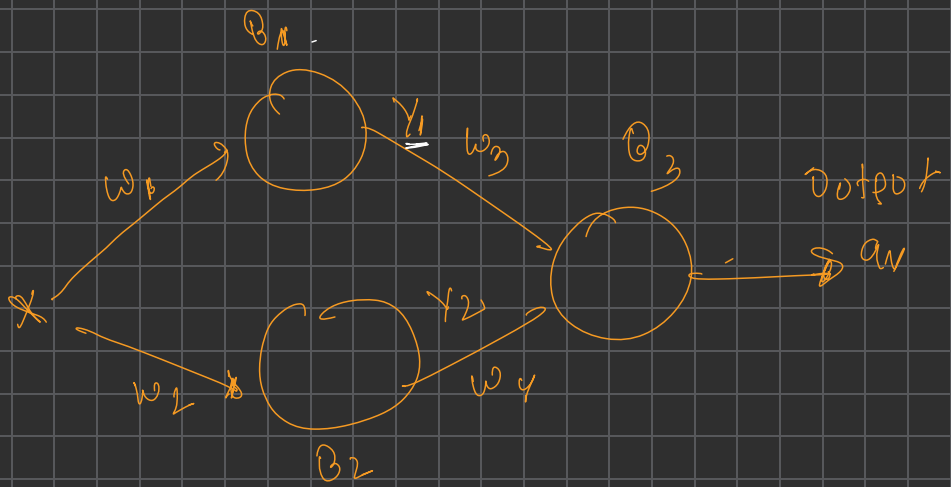


$$\text{Output} = w_2 * Y + \theta_2$$

$$\Rightarrow w_2 * (w_1 * x + \theta_1) + \theta_2$$

$$\Rightarrow w_1 * w_2 x + w_2 * \theta_1 + \theta_2$$

$$\text{Output} \Rightarrow mx + c$$



Output $\Rightarrow$ $Y_1 * w_3 + Y_2 * w_4 + \theta_3$

$= (w_1 * x + \theta_1) * w_3 + (w_2 * x + \theta_2) * w_4 + \theta_3$

$\Rightarrow w_1 * w_3 x + w_3 \theta_1 + w_2 * w_4 x + w_4 * \theta_2 + \theta_4$

$\Rightarrow x(w_1 * w_3 + w_2 * w_4) + C$

$$\boxed{\text{Output} \Rightarrow mx + C}$$

16 ? 2 ans

$16 \times 16 \times 16 = 4$

X	Y
2	9
2	3
4	8
5	12
8	17

1	1
2	8
3	27
4	64
5	125

Activation Function

$y \le 0$, output

$y > 0 \Rightarrow y$

Activation

RELU $\Rightarrow$

X	Y
1	0
2	0
3	0
4	1
5	2
6	3
7	4
8	5

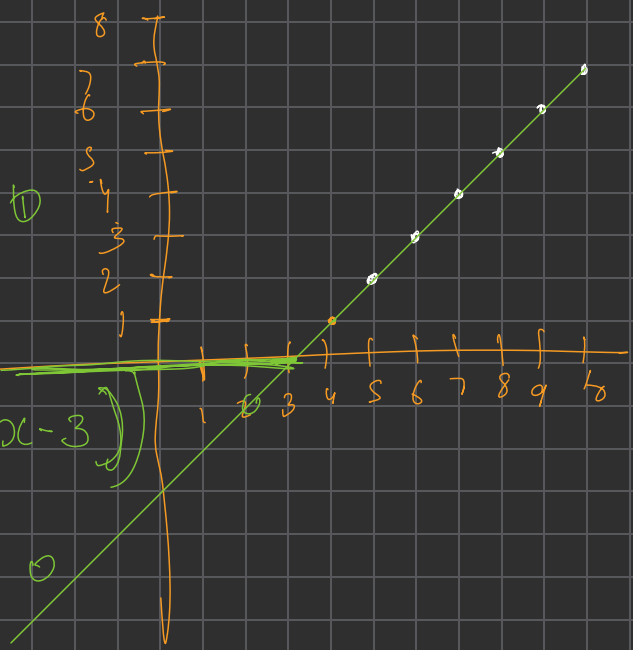
$4-3$

$1 \to 0$

$(\text{RELU}(x-3))$

$x-3$

$-2 \quad 0$



$+\theta$

w_1

x

Activation fn.

$\Rightarrow (w_0 x + \theta)$ RELU

$w_1 x + \theta <= 0 \to 0$

$w_1 x + \theta > 0 \to 1$

x

$\Rightarrow w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + \theta$

1 — 1

2 — 4

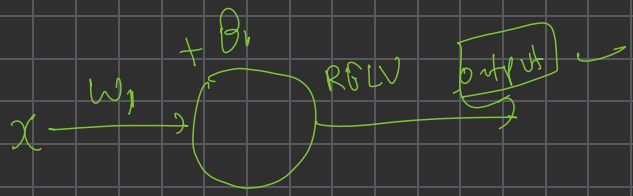
3 — 9

4 — 16

5 — 25

6 — 36

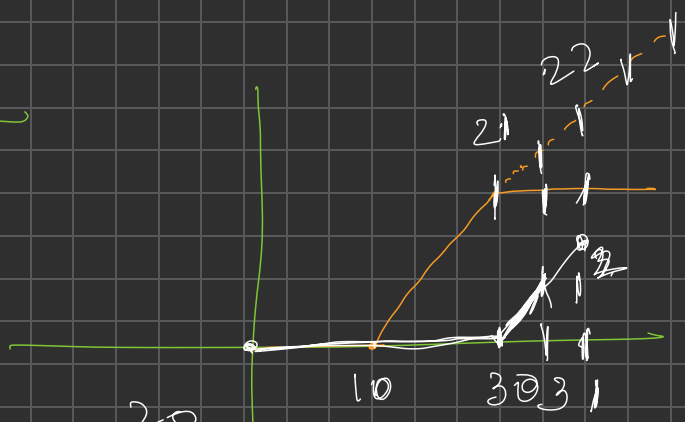
$$\text{Output} = w_1 x + w_2 y + w_3 z + w_4 k + \cdots$$



$$\text{RELU}(w_1 \cdot x + \theta)$$

$$\text{RELU}(x - 3)$$

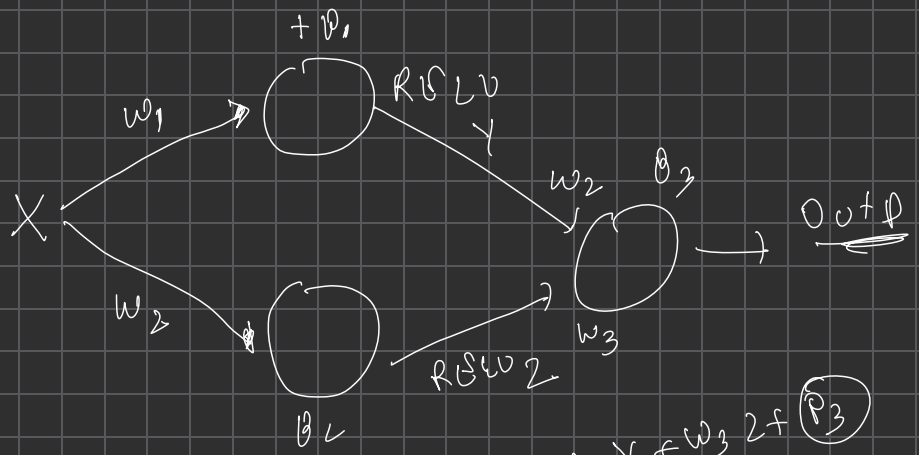
X	Y
0	0
1	0
2	0
7	
9	0
10	0
11	1
12	2
13	3
30	20
31	20
32	20
33	20
34	20
35	20



$$\boxed{RELU(x-10) - RELU(x-30)}$$

$25 - 5$

$\Rightarrow 20$



$$\text{Output} = w_2 Y + w_3 Z + \theta_3$$

$$= Y - Z$$

$$= \text{RELU}(X-10) - \text{RELU}(X-30)$$